

TASK 7

DDoS Attack Investigation Report

Incident: AWS 2.3 Tbps DDoS Attack (February 2020)

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1. Target

The attack was mitigated by Amazon Web Services (AWS) in February 2020. AWS did not reveal the specific customer targeted, but the malicious traffic was directed towards an application hosted on AWS Shield Advanced, a managed DDoS protection service.

2. Technology Used

- Attack peak: 2.3 Terabits per second (Tbps).
- Duration: Several minutes which was detected and neutralized by AWS Shield within a few seconds.
- Attack type: Reflection-based volumetric DDoS using CLDAP (Connection-less LDAP) amplification.
- Amplification factor: Roughly 56-70x, making small requests trigger large responses toward the victim.
- Likely sources: Botnets exploiting misconfigured CLDAP servers on the internet.
- Main effect: Flooded target infrastructure with massive

traffic volumes which could have been attempting to saturate network capacity.

3. Attacker's Motive

- No public attribution.
- Possible motives include:
 - Testing newly discovered reflection vectors.
 - Disrupting a high-value AWS-hosted service.
 - Demonstrating technical capability for underground reputation.
 - Distraction from other malicious activity or ransom attempts.

4. Overall Impact

- AWS Shield mitigated the attack without customer impact.
- However, it revealed how reflection-based amplification attacks could achieve multi-Terabit scale.
- Key insights:
 - Legacy internet protocols like CLDAP and Memcached remain dangerously exploitable.
 - Large-scale mitigation infrastructure is essential for cloud providers.
 - Smaller networks would likely collapse under such load.

5. Defensive Strategies

1. Managed DDoS Protection (AWS Shield, Cloudflare, Akamai) - absorbs volumetric floods at scale.
2. Protocol hardening - disable or secure exposed CLDAP, NTP, and Memcached servers.
3. Automated anomaly detection - use traffic baselines to detect sudden volumetric deviations.
4. Ingress and egress filtering (BCP38) - prevent spoofed traffic amplification.
5. Global monitoring networks - identify emerging reflection vectors early.
6. Collaboration with ISPs and CERTs - coordinate to shut down misconfigured amplification sources.
7. Application-layer throttling - ensure systems fail gracefully under stress.

Conclusion

The AWS 2.3 Tbps CLDAP reflection attack showcased how amplification techniques can be weaponized to reach multi-terabit scales. While AWS absorbed the impact, the event was a wake-up call for smaller organizations relying solely on perimeter defenses. Modern DDoS defense requires a multi-layered approach: global traffic scrubbing, automated mitigation, and securing legacy protocols. The attack reinforced that internet resilience depends on

global cooperation and proactive hygiene-not just large-scale infrastructure.